DEEP LEARNING ASSIGNMENT-2 Report

Title: *Environmental Sound Classification Using CNN and MFCCs*

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M.Tech (Robotics and AI), 2nd Semester

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1. Problem Statement:

To build a sound classification system that detects **10 nature sounds** (e.g., Rainfall, Thunder, Fire, Wind, etc.) using deep learning. The goal is to automatically recognize the type of sound from short .wav/.mp3 audio clips.

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2. Methodology:

- **Dataset:** 100 audio files across 10 nature classes (Rain, Thunder, Ocean, etc.)
- **Preprocessing:**
  - Loaded audio using **Librosa**
  - Standardized duration to **4 seconds**
  - Extracted features: raw waveform & later **MFCCs (Mel-Frequency Cepstral Coefficients)**
- **Model 1:** 1D CNN on raw audio (Accuracy ≈ **30%**)
- **Model 2:** 2D CNN on **MFCCs** (Improved Accuracy up to **50%**)
- **Frameworks:** TensorFlow, Keras, Google Colab
- **GUI:** Implemented **Gradio** interface to test live audio samples

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3. Results:

Model	Final Test Accuracy	Loss
1D CNN (Raw)	30.00%	2.5045
2D CNN (MFCC)	**Up to 50.00%**	2.0784

- **Best Validation Accuracy:** 50%

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1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 128ms/step - accuracy: 0.5000 - loss: 2.0784
✅ Final Test Accuracy: 50.00%
❌ Final Test Loss: 2.0784
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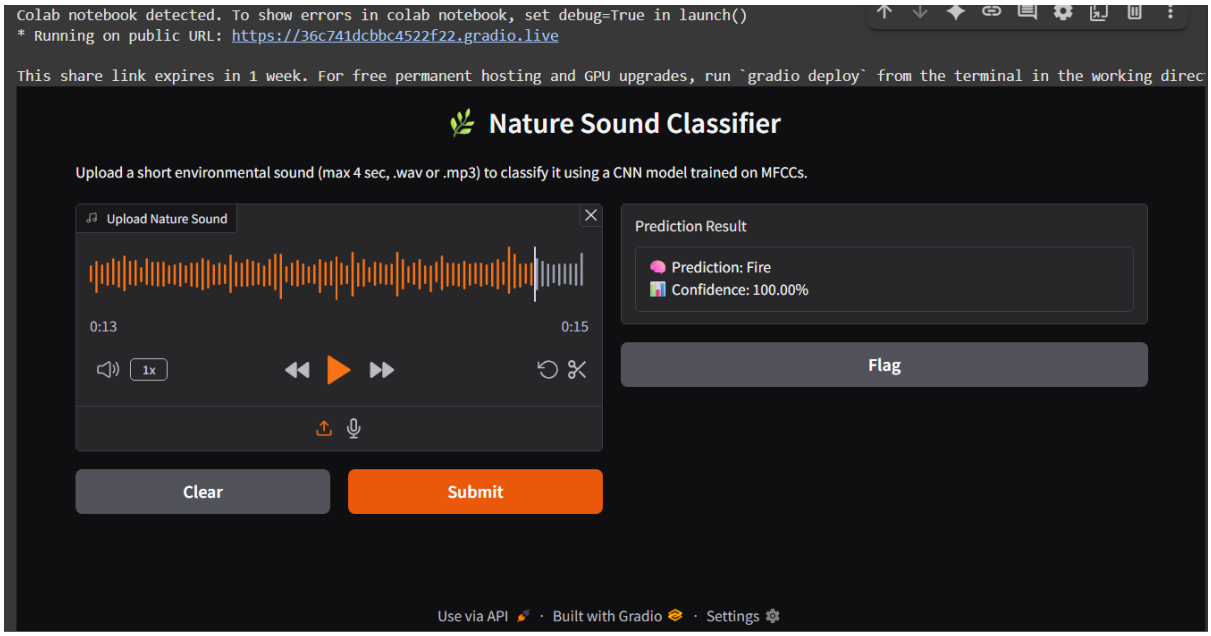
- **Classified Sounds:** Wind, Thunder, Rainfall, Ocean, Forest, Night, Fire, City, Elephants, Firecrackers



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4. GUI Link (Gradio):

Kaggle Login Details:

Email - rai.deeplearning.kaggle@gmail.com
Password - raidl123

Google Colab Link:

Assignment-2 (Sound Classification Project):
https://colab.research.google.com/drive/1SMnQqEAbcaVjbiYMUi4kx7VK0igZ0Zsc?usp=sharing

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5. Conclusion:

- MFCCs significantly improved classification accuracy.
- The model performs reasonably well for environmental monitoring or smart sound recognition.
- Future work: Add background noise filtering and data augmentation for real-world accuracy.